Vision-Based Autonomous Cleaning Robot

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***Abstract*—This report presents the mid-term progress of a vision-based autonomous cleaning robot designed to detect waste objects, navigate toward them, and dispose of them in a dustbin. The work extends the problem of dynamic path planning of mobile robots with real-time sensor feedback. The current implementation focuses on simulation using ROS/ROS2 and Gazebo, along with a potential field-based navigation framework for obstacle avoidance. Preliminary results demonstrate successful goal-reaching behavior in static environments. Future work aims to incorporate dynamic obstacles, real-time sensor integration (LiDAR and camera), and vision-based object detection to achieve complete autonomous cleaning behavior.**

I. Introduction

Autonomous mobile robots are increasingly required to operate in dynamic and unstructured environments, particularly in human-centric spaces. Traditional navigation systems rely on sensors such as LiDAR and ultrasonic sensors for obstacle avoidance but lack the capability to intelligently interact with objects in the environment.

This project is inspired by the problem of *path planning of a mobile robot with real-time feedback from mounted camera and LiDAR sensors*, where dynamic environments require adaptive and responsive navigation strategies. In such scenarios, classical approaches like potential field methods must be enhanced or replaced with more robust planning and control techniques.

The objective of this work is to extend this problem into a task-oriented system by developing a vision-based autonomous cleaning robot. The robot should not only navigate safely but also detect waste objects, move toward them, and dispose of them in a designated location.

II. Planned Methodology

The proposed system is structured into multiple modules:

Perception: A camera-based perception system will be implemented using OpenCV and deep learning models such as YOLO to detect waste objects like paper wrappers.

Sensor Integration: LiDAR data will be incorporated for accurate environment perception and obstacle detection in real time.

Localization: The position of detected objects will be estimated relative to the robot using sensor data and simulation coordinates.

Path Planning: Initially, potential field-based methods are used. Further exploration will include advanced algorithms such as A*, D*, and sampling-based planners. Learning-based approaches such as Reinforcement Learning and Model Predictive Control (MPC) will also be investigated for dynamic environments.

Control and Navigation: The robot will continuously adjust its trajectory based on real-time sensor feedback.

Action Module: A mechanism will be simulated for picking up detected waste and disposing it in a predefined dustbin.

Simulation Environment: The entire system is implemented using ROS/ROS2 and Gazebo to emulate realistic robotic behavior.

III. Work Completed

The following progress has been achieved:

- Explored and configured ROS and ROS2 environments for robotic development.
- Set up Gazebo simulation for a mobile robot platform.
- Implemented a potential field-based path planning algorithm.
- Enabled robot navigation from a start position to a goal position.
- Incorporated static obstacles into the simulation environment.
- Developed a basic visualization interface to observe robot movement.
- Conducted multiple experiments to validate navigation performance.
- Uploaded implementation code to a GitHub repository.

GitHub Repository:
https://github.com/Nitesh0409/autonomous_cleaning_robot

IV. Preliminary Results and Discussion

The implemented system demonstrates successful navigation using a potential field-based approach. The robot is able to reach the specified goal while avoiding static obstacles in the environment.

However, several limitations were observed:

- Sensitivity to goal position changes.
- Occurrence of local minima, causing inefficient or stuck behavior.
- Lack of adaptability to dynamic obstacles.

These challenges highlight the need for more advanced planning and control strategies, particularly in dynamic environments.

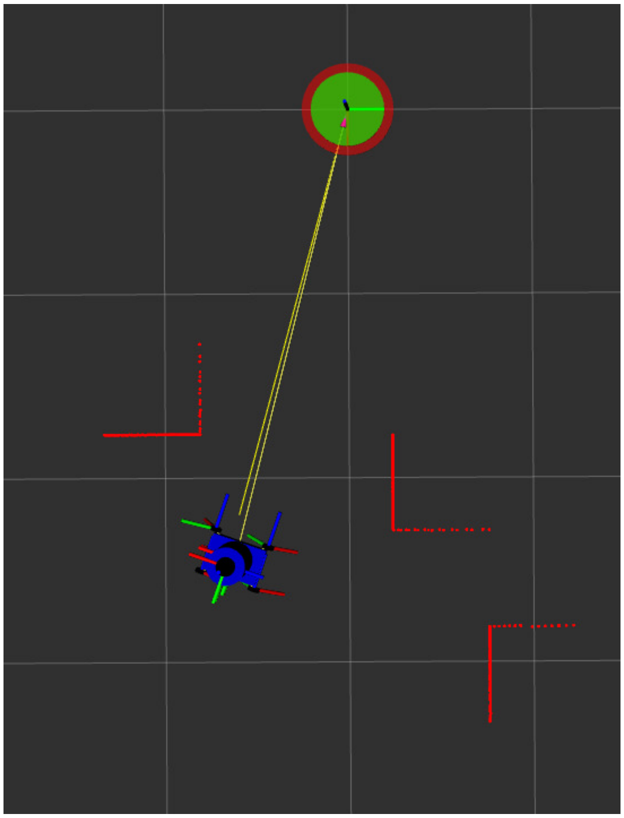

Fig. 1. Navigation using potential field method

V. Plan for Pending Work

The next phase of the project will focus on extending the system toward a complete autonomous cleaning pipeline:

- Exploration and implementation of advanced path planning algorithms such as A*, D*, and hybrid approaches.
- Extension of the simulation environment to include dynamic (moving) obstacles.
- Integration of LiDAR data for real-time obstacle detection and mapping.
- Development of a vision-based object detection module using YOLO and OpenCV.
- Fusion of camera and LiDAR data for improved perception.
- Implementation of a pick-and-place mechanism for waste collection.
- End-to-end system integration: detection $\rightarrow$ navigation $\rightarrow$ collection $\rightarrow$ disposal.
- Investigation of learning-based approaches such as Reinforcement Learning and Model Predictive Control for adaptive navigation.

VI. Work Done by Each Member

Both members actively contributed to the exploration and development of the system. Initial work involved independent experimentation with ROS, ROS2, and Gazebo to understand system behavior. The best-performing implementations were selected collaboratively.

As the project progresses, responsibilities will be more clearly divided across modules such as perception, planning, and system integration.

VII. References

References

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